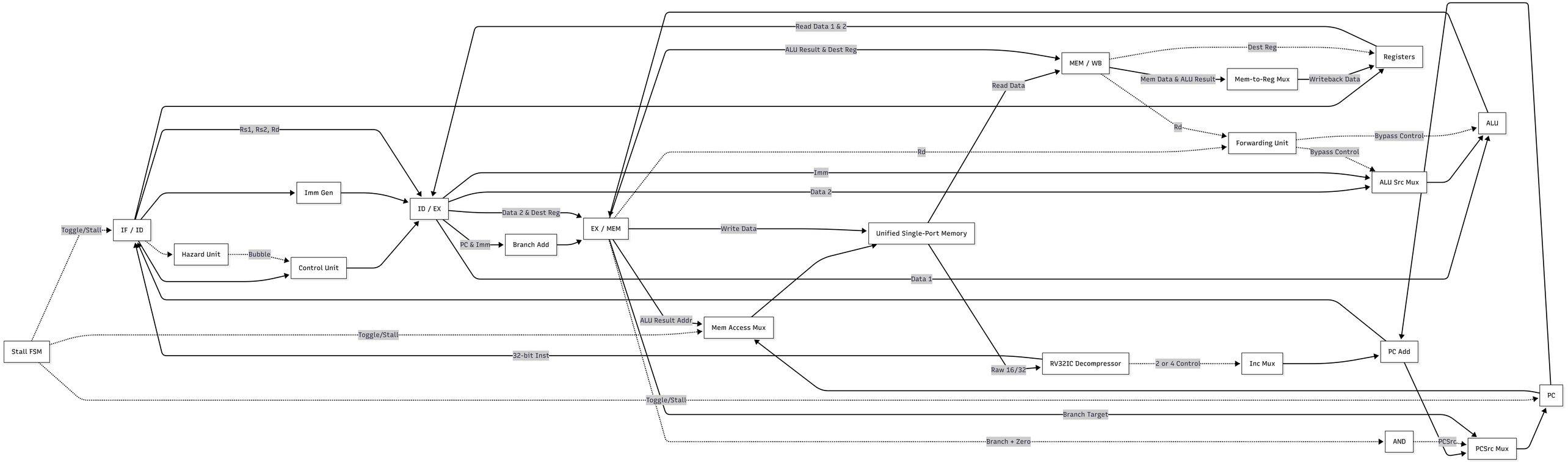
Read Data 1 & 2
ALU Result & Dest Reg
MEM / WB
Dest Reg
Registers
Mem Data & ALU Result
Mem-to-Reg Mux
Writeback Data
Read Data
Rd
Forwarding Unit
Bypass Control
ALU
Bypass Control
Rs1, Rs2, Rd
Rd
Imm
ALU Src Mux
Imm Gen
Data 2
ID / EX
Data 2 & Dest Reg
EX / MEM
Write Data
Unified Single-Port Memory
Toggle/Stall
IF / ID
PC & Imm
Branch Add
Hazard Unit
Bubble
Control Unit
Data 1
ALU Result Addr
Mem Access Mux
Toggle/Stall
Stall FSM
32-bit Inst
RV32IC Decompressor
2 or 4 Control
Inc Mux
PC Add
Raw 16/32
PC
Toggle/Stall
Branch Target
Branch + Zero
AND
PCSrc
PCSrc Mux